UnJin Lee, PhD

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Positions

2024-	Postdoctoral Fellow, Zhao Lab, The Rockefeller University *National Science Foundation Postdoctoral Research Fellowship in Biology*
2022-2024	Postdoctoral Associate, Zhao Lab, The Rockefeller University

Education

PhD, MS University of Chicago
Committee on Genetics, Genomics, and Systems Biology, 12/2021
On the Mechanisms of Genomic and Phenotypic Evolution
Advisor: Manyuan Long

BA, University of Chicago
Physics, 2013

Research Experience

Publications

U. Lee†, L. Zhao†, HP6/Umbrea, a rapidly evolving Drosophila HP1-family paralog, is a candidate HP1a-recruited plasticizer of heterochromatin. (under review: *Biophysical Journal*) preprint

†*corresponding author* *Biomolecular condensates, heterochromatin, molecular dynamics*

U. Lee†, Naturally arising de novo open reading frames as potential zinc chelators in Drosophila melanogaster. (in press: *Genome Biology and Evolution*) preprint

†*corresponding author* *transcriptional noise, metalloproteins, microsatellites, de novo gene origination*

S. Mozeika, **U. Lee**, S. Mills, N. Svetec, L. Zhao, Evolution of behavioral dynamics underlying mechanosensory oviposition preference in D. suzukii. (submitted)

C. Thompson, S. Mozeika, E. Paredes, **U. Lee**†, Korean Natural Farming practices are dominated by a limited number of microbes and decrease fungal diversity. *Sustainable Microbiology*, 3(3):qvag033 (2026) pubmed

†*corresponding author* *public outreach, metagenomic sequencing, microbial dynamics*

N. Svetec, **U. Lee**, L. Zhao, Machine learning for evolutionary genetics and molecular evolution. *Trends in Genetics*, 2026Mar24:S0168-9525(26)00032-6 (2026) pubmed

U. Lee, C. Li, C. B. Langer, N. Svetec, L. Zhao, Comparative single cell analysis of transcriptional bursting reveals the role of genome organization on de novo transcript origination. *PNAS,* **122**(18):e2425618122 (2025) pubmed

single-cell transcriptomics, genome organization, statistical methods, transcriptional bursting, de novo origination

U. Lee†*, D. Arsala*, S. Xia*, C. Li, M. Ali, N. Svetec, C.B. Langer, D. Sobreira, I. Eres, D. Sosa, J. Chen, L.Zhang, P. Reilly, A. Guzzetta, J.J. Emerson, P. Andolfatto, Q. Zhou, L. Zhao, M. Long†. The 3-Dimensional Genome Drives the Evolution of Asymmetric Gene Duplicates via Enhancer Capture-Divergence. *Science Advances,* **10**, eadn6625 (2024) pubmed

†*corresponding author* *HiC-Seq, RNAi, genome organization, gene regulation, gene duplication, evolutionary theory*

U. Lee, S. Mozeika, L. Zhao, A Synergistic, Cultivator Model of De Novo Gene Origination. *Genome Biology and Evolution*. evae103 (2024). pubmed

evolvability, non-coding RNA, de novo origination, evolutionary theory

U. Lee†, E. N. Mortola, E. Kim, M. Long. Evolution and maintenance of phenotypic plasticity. *BioSystems*. **222**, 104791 (2022). pubmed

†*corresponding author* *phenotypic plasticity, stochastic differential equations, evolutionary theory*

S. Xia, N. W. VanKuren, C. Chen, L. Zhang, C. Kemkemer, Y. Shao, H. Jia, **U. Lee**, A. S. Advani, A. Gschwend, M. D. Vibranovski, S. Chen, Y. E. Zhang, M. Long, Genomic analyses of new genes and their phenotypic effects reveal rapid evolution of essential functions in Drosophila development. *PLOS Genetics*. **17**, e1009654 (2021). pubmed

J. Zu, Y. Gu, Y. Li, C. Li, W. Zhang, Y. Zhang, **U. Lee**, L. Zhang, M. Long, Topological evolution of coexpression networks by new gene integration maintains the hierarchical and modular structures in human ancestors. *Science China Life Sciences*. **62** (2019). pubmed

E. Leypunskiy, J. Lin, H. Yoo, **U. Lee**, A. R. Dinner, M. J. Rust, The cyanobacterial circadian clock follows midday in vivo and in vitro. *eLife*. **6**, e23539 (2017). pubmed

E. Kim, **U. Lee**, J. Heseltine, R. Hollerbach, Geometric structure and geodesic in a solvable model of nonequilibrium process. *Phys. Rev. E*. **93**, 062127 (2016). pubmed

U. Lee†, J. J. Skinner, J. Reinitz, M. R. Rosner, E.-J. Kim†, Noise-driven phenotypic heterogeneity with finite correlation time in clonal populations. *PLOS ONE*. **10**, e0132397 (2015). pubmed

†*corresponding author* *phenotypic plasticity, stochastic differential equations, evolution of treatment resistance*

U. Lee*, C. Frankenberger*, J. Yun, E. Bevilacqua, C. Caldas, S.-F. Chin, O. M. Rueda, J. Reinitz, M. R. Rosner, A prognostic gene signature for metastasis-free survival of triple negative breast cancer patients. *PLOS ONE*. **8**, e82125 (2013). pubmed

machine learning, triple-negative breast cancer, patient prognosis, survival analysis

*contributed equally

†corresponding author

Intellectual Property

Compositions and methods for snap freezing and cryoembedding of biological samples (**U. Lee**, L. Zhao, USPTO Prov. Patent filed: 08/31/2026)

methodology and device related to microscopic tissue preparation for cryosectioning, spatialomics, etc.

Prognostic and predictive breast cancer signature (M. Rosner, M. Sun, **U. Lee,** USPTO 14/952,265)

gene expression-based prognostic signature for the most aggressive breast cancers

Presentations

2026 Population, Evolutionary, and Quantitative Genetics Conference
Pacific Grove, CA
Comparative single-cell analysis of transcriptional bursting reveals the role of genome organization in de novo transcript origination (poster)

2025 University of Oklahoma Health Sciences Center
Oklahoma City, OK
Where do new genes come from? How genome structure drives new gene origination (invited talk)

2025 66th Annual Drosophila Research Conference
Single-cell research in Drosophila: technologies and resources workshop
San Diego, CA
scHSQ allows for the alignment of evolutionarily diverged tissue types (invited workshop talk)

2024 NY Area Population Genetics Meeting
New York City, NY

Comparative scRNA-Seq analysis of Drosophila spermatogenesis reveals critical role of genome organization in de novo transcript origination (selected talk)

2023 SMBE Satellite Meeting on De Novo Genes
College Station, TX
*Origination of a Testis-Specific lncRNA (poster, **best presentation award**)*

2022 NY Area Population Genetics Meeting
New York City, NY
Evolution and Maintenance of Phenotypic Plasticity (poster)
The Role of the 3D Genome in New Gene Evolution (poster)

2019 Midwest Population Genetics Meeting
Chicago, IL
Evolutionary Consequences of Phenotypic Diversification (selected talk)

2019 Gordon Conference on Ecological and Evolutionary Genomics
Manchester, NH
Regulatory Evol. of a New Essential Gene is Driven by an Old Enhancer (poster)

2016 Aspen Center for Physics on Evolution, Populations, and Physics
Aspen, CO
Selection Over a Model of Noise-Driven Phenotypic Heterogeneity (poster)

2013 American Association of Cancer Researchers Annual Conference
Washington, DC
A Prognostic Gene Sig. for Metastasis-Free Survival of TNBC Patients (poster)

Awards and Honors

2026 Black Family Therapeutic Development Fund Proof-of-Concept Award
Co-PI $75,000

2024-2027 NSF Postdoctoral Research Fellowship in Biology Recipient
PI $240,000

2023 Best Presentation Award, SMBE Satellite Meeting on De Novo Genes

2023 SMBE Satellite Meeting on De Novo Genes Travel Award

2017-2021 Ridgeway Endowment Support

2017 Hinds Evolutionary Biology Graduate Student Research Award

2016 NSF Graduate Research Fellowship Program Honorable Mention

2015-2017 NIH T32 Gene Regulation Training Grant Recipient

2011 NSF Res. Experiences for Undergraduates, Chicago Center for Systems Biology

2008-2013 Odyssey Scholarship Recipient, University of Chicago

Service Activities

Outreach

Summ. 2023 RockEDU SSRP Mentor

Spring 2023 Genspace FIT Scholar Program Mentor

2017-2019 EE Just Portrait Commission

2016-2018 Co-Host Groks Science Radio Hour WHPK 88.5FM

2015-2016 Genetics Science Connections at Museum of Science and Industry

Professional

2023- Rockefeller University Sustainability Committee Laboratory Representative

2023-2024 Rockefeller University Post-Doc Association Board Member

2022- Rockefeller Inclusive Science Initiative Executive Board Member (Mental Health and Accessibility Coordinator 2023-)

2017-2018 BSD Equipment Library Committee (Founder)
2017-2019 BSD Travel Award Committee (Member, Chair 2018-2019)
2014-2019 Dean's Council Representative
2014-2016 Program Representative (GGSB)

Teaching

Instruction

Fall 2023 Workshop Instructor, Genspace ("*Introduction to Bioinformatics*")
Spring 2023 Workshop Instructor, RockEDU SSRP ("*Introduction to Molecular Evolution*"; "*Detecting Signatures of Selection*"; "*Genetic Sources of Phenotypic Diversity*")
Spring 2023 Workshop Instructor, Genspace ("*Introduction to Bioinformatics*")
Winter 2018 Grader, University of Chicago ("*Fundamentals of Genetics*")
Spring 2017 Teaching Assistant, University of Chicago ("*Classics of Evolutionary Genetics*")
Winter 2017 Grader, University of Chicago ("*Fundamentals of Genetics*")
Fall 2016 Teaching Assistant, University of Chicago ("*Developmental Biology*")

Trainees

2024- Candace Thompson (*"Microbial Community Propagation Through KNF Practices Decreases Fungal Diversity and Increases Microbial Diversity")*
Summ. 2023 Yifei Zhou ("*Ablation & Genesis: Comparative Genetic Analysis of Behavioral Preference in an Invasive Pest Species*"
Summ. 2023 Finn Charest ("*Randomness is the Spice of Life: Examining Stochastic Fluctuations in Lotka-Volterra Models*")
Summ. 2023 Bharat Elango ("*Machine Learning-Based Analysis of Gene Expression in D. melanogaster*")
Spring 2023 Kiera Derrig ("*Curating My Arsenal of Colorants with Bacteria*")
Summ. 2019 Gustavo Rangel T. de Almeida ("*Numerical Optimization Methods in Python*")
2017-2019 Jack Mortola ("*Regulatory Evolution of a New Essential Gene is Driven by an Old Enhancer*")

Professional Skills Development

Summ. 2025 Academic Job Search Boot Camp
Participant
Summ. 2024 Tri-Institutional Transitioning to Research Independence Series
Participant
Summ. 2024 Managing Science and Scientists Workshop
Invited Alumni Panelist
Summ. 2023 Managing Science and Scientists Workshop
Three day management workshop taught by Harvard Business School and Haas School of Business faculty
Workshop Participant (Inaugural)

Scientific Software

scHSQ (R package) *companion method for Lee, et al. 2025*
sigsquared (Bioconductor/R package) *companion method for Lee, et al. 2013*

Peer Review

PLoS Computational Biology, Cancer Cell International (Springer), Genome, National Science Foundation (*ad hoc* reviewer), PNAS, Journal of Evolutionary Biology, Journal of Molecular Evolution, Genome Biology and Evolution, Molecular Biology and Evolution